

Maxence Noble

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WORK EXPERIENCE

CFM ML Lab <i>Paris, France</i>	Postdoctoral Researcher <i>Supervisor : Eric Vanden-Eijnden</i> Working on fundamental principles and applications of generative models and sampling methods. Collaborating with ENS Ulm (Center for Data Science).	Sep. 2026- now
Jasper AI (Image team) <i>Paris, France</i>	Research Scientist Intern <i>Supervisor : Clément Chadebec</i> Worked within the Research team on distillation of diffusion-based generative models applied to image super-resolution tasks [a].	Mar. 2025- Sep. 2025
ENS Ulm (Center for Data Science) <i>Paris, France</i>	Graduate Research Assistant <i>Supervisor : Valentin de Bortoli</i> Worked on sampling algorithms for constrained spaces with self-concordant barriers [3].	Apr. 2022- Sep. 2022
Ecole Polytechnique (CMAP) & INRIA Lille <i>Palaiseau/Lille, France</i>	Graduate Research Assistant <i>Supervisors : Aymeric Dieuleveut, Aurélien Bellet</i> Worked on federated learning comprising concerns about privacy and data heterogeneity [1].	Apr. 2021- Nov. 2021
Deepki <i>Paris, France</i>	Data Scientist Intern Worked within the R&D team on improving data completeness in the collection and use of energy consumption invoices (scraping & parsing).	Jun. 2020- Aug. 2020

EDUCATION

Ecole Polytechnique (CMAP) <i>Palaiseau, France</i>	Ph.D. in Machine Learning <i>Supervisor : Alain Durmus</i> Thesis: “Sampling and Transport with Score-based Diffusion models: Opportunities and Methodological Advances” Papers : [2-8], [b-c]	Sep. 2022- Jun. 2026
ENS Paris-Saclay <i>Gif-sur-Yvette, France</i>	M.Res. in Machine Learning (MVA) <i>Distinctions with jury’s congratulations</i>	Sep. 2021- Sep. 2022
Ecole Polytechnique <i>Palaiseau, France</i>	M.Sc. in Applied Maths (Ingénieur Polytechnicien) <i>GPA 3.93/4.00, top 10% of all students</i>	Aug. 2018- Aug. 2021
University of Nanterre <i>Nanterre, France</i>	B.A. in Philosophy (Add. distance learning degree) <i>Honours</i>	Oct. 2019- Jul. 2020
Sainte Geneviève college <i>Versailles, France</i>	French preparatory classes <i>GPA 3.97/4.00</i>	Aug. 2016- Aug. 2018

PREPRINTS

[c] Twisted Schrödinger Bridge Matching, MN, Marie Scheid, Yazid Janati, Eric Moulines, Alain Durmus	2026 (<i>in review</i>)
[b] Sampling from multi-modal distributions on Riemannian manifolds with training-free stochastic interpolants, A. Durmus, MN, T. Pellerin	2026 (<i>in review</i>)
[a] Fast, faithful and photorealistic diffusion-based image super-resolution with enhanced Flow Map models, MN, G. Inkana Quintana, C. Chadebec, B. Aubin	2026 (<i>tech. report</i>)

PUBLICATIONS

[8] Diffusion-based Annealed Boltzmann Generators : benefits, pitfalls and hopes, L. Grenioux*, MN*	TMLR 2026
[7] Improving the evaluation of samplers on multi-modal targets, L. Grenioux*, MN*, M. Gabrié	ICLR 2025 (<i>FPI workshop</i>)
[6] Learned Reference-based Diffusion sampling for multi-modal distributions, MN*, L. Grenioux*, M. Gabrié, A. Durmus	ICLR 2025
[5] Stochastic Localization via Iterative Posterior Sampling, L. Grenioux*, MN*, M. Gabrié, A. Durmus	ICML 2024 (<i>spotlight</i>)
[4] Tree-based Diffusion Schrödinger Bridge with Applications to Wasserstein Barycenters, MN, V. de Bortoli, A. Doucet, A. Durmus	NeurIPS 2023 (<i>spotlight</i>)
[3] Unbiased constrained sampling with self-concordant Barrier Hamiltonian Monte Carlo, MN, V. De Bortoli, A. Durmus	NeurIPS 2023
[2] Non-asymptotic convergence bounds for Sinkhorn iterates and their gradients: a coupling approach, G. Greco, MN, A. Durmus, G. Conforti	COLT 2023
[1] Differentially Private Federated Learning on Heterogeneous data, MN, A. Bellet, A. Dieuleveut	AISTATS 2022

TEACHING EXPERIENCE

Teaching Assistant

- MRe ‘MVA’ – Ecole Normale Supérieure Paris-Saclay
 - Computational Statistics
- MSc Data Science for Business - Ecole Polytechnique & HEC Paris
 - Machine Learning, Statistics, Optimization, and Linear Algebra
- BSc in Mathematics – Ecole Polytechnique

Sep. 2022-
Sep. 2024

REVIEWING EXPERIENCE

NeurIPS, NeurIPS PriGM workshop, TMLR	2026
ICLR, ICLR FPI workshop, NeurIPS	2025
ICML (<i>Top reviewer</i>), NeurIPS, TMLR	2024
AISTATS, UAI, ICML SPIGM workshop, NeurIPS (<i>Top reviewer</i>), TMLR	2023
NeurIPS	2022

SCIENTIFIC EVENT ORGANIZATION

NeurIPS@Paris workshop (*co-organizer*)

2024, 2025

TALKS

Center for Data Science SALT reading group, ENS Paris [6]	February 2025
Google DeepMind's workshop on diffusion models, Paris [6]	January 2025
Isaac Newton Institute for Mathematical Sciences Workshop, Alan Turing Institute [5]	July 2024
4-th Italian Meeting on Probability and Statistics, Rome [3]	June 2024
Google DeepMind's reading group on generative models, transport and sampling [5]	May 2024
Mostly Monte Carlo Seminar, Paris Santé Campus [5]	April 2024
ELLIS un-conference ICML, HEC Paris [3]	July 2023
Isaac Newton Institute for Mathematical Sciences Workshop, Cambridge University [4]	June 2023

AWARDS AND HONOURS

- 2022: Awarded a selective PhD scholarship by Ecole Polytechnique.
- 2021: Awarded the Ecole Polytechnique Research Centre Prize in Applied Mathematics for [1].
- 2021: Outstanding Investment Award for involvement in the Polytechnique student community.
- 2019: National defence medal (bronze level).

SKILLS

Programming	Python, PyTorch, Git, Slurm
Languages	French (native) - English (fluent) - German (advanced) - Italian (proficient)